Hyung Cho

Software Engineer | Agentic Integrations, LLM Evals, and Test Automation

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SUMMARY

Builds LLM integrations and the evals that keep them honest: a tool-calling agent loop on the OpenAI API, retrieval and escalation evals that run on pull requests, and a failing LLM-judge gate traced to its root cause. Ships with AI coding agents under a written risk assessment on each change, backed by 10 years of validation engineering at Zebra Technologies, including C# test tooling and SDK validation.

EXPERIENCE

Independent AI Engineer | Self-directed projects, Remote **Jun 2025 - Present**

Solo product work, built by directing AI coding agents (Claude Code, Codex).

Purvia, permission-aware RAG and support-bot app, open source

- Built a framework-free tool-calling agent loop on the OpenAI Chat Completions API with 4 tools (document search, web search, text-to-SQL, and a document sub-agent), capped at 5 iterations.
- Built a 60-question retrieval test set scoring vector, keyword, and hybrid search on recall@k, MRR, and nDCG@5; a PR workflow posts the score change as a comment and flagged a -0.510 recall@5 drop on a deliberately broken PR.
- Designed an escalation eval whose deterministic leak checks block merges and whose LLM-judged checks run on a schedule against a 5% false-resolve ceiling.
- Traced that eval's four-run breach to the offline LLM judge, not the product, then compared five judge prompt and schema designs on replayed drafts and cut wrong verdicts from 38 of 280 to 0 of 300.
- Tightened the live answer gate's judge rubric after measuring it pass a wrong-value reply 10 of 20 times; the shipped rubric sent 1 of 66 should-escalate drafts where a stronger model sent 15 of 66.

Rep Today, native iOS workout app, pre-release

- Integrated OpenAI through a Cloudflare Worker proxy (store: false, capped output, 30-second timeout) with App Attest and subscription checks on the server; 477 proxy Vitest tests pass in CI.
- Directed development as 214 agent-authored pull requests, 195 with a written risk assessment; later ones separate scenarios validated live in the app from those covered only by tests.

CreativeBridge, K-12 AI story-writing iOS app, internal TestFlight builds

- Integrated OpenAI GPT-4o-mini story writing through Convex server actions and Replicate image generation that falls back from a Flux model to Stable Diffusion 3.5 on error or timeout.

Volunteer AI/ML Engineer | Omdena, Remote **Apr 2023 - Jun 2025**

Unpaid collaborative ML projects with global volunteer teams.

- Fine-tuned LLaMA and Falcon, integrated ChatGPT into a chatbot, and iterated on prompts with an evaluation workflow.

Career break and ML retraining **Aug 2018 - Mar 2023**

Self-directed ML study and the FourthBrain ML Engineering bootcamp (2022); 29 public ML projects on GitHub.

Senior Validation Engineer | Zebra Technologies, Agoura Hills, CA **Feb 2008 - Jul 2018**

- Led validation testing for 6 shipped hardware products, writing test plans, procedures, acceptance criteria, and final reports.
- Built C# validation tooling that replaced manual test workflows and cut regression test cycles by about 30%.
- Isolated root causes of failures across hardware, firmware, software, and test infrastructure, and proved each fix.
- Mentored 10 test technicians and set quality and lab process standards adopted across the products the team validated.

SKILLS

LLM integration and agents: OpenAI API, Anthropic API, Tool calling, Agent loops, Prompt engineering, RAG
Evals and testing: LLM evaluation, LLM-as-judge, Retrieval metrics, Regression testing, GitHub Actions, CI/CD
Languages and platforms: Python, FastAPI, TypeScript, JavaScript, PostgreSQL, Cloudflare Workers, Docker
AI-assisted engineering: AI coding agents, Claude Code, Code review, Technical writing, Mentoring

EDUCATION AND TRAINING

B.S., Electrical Engineering, California Polytechnic State University, San Luis Obispo
FourthBrain ML Engineering bootcamp (2022); DeepLearning.AI deep learning and generative AI courses